Notes: Phillips (JFE, 1995)

Increased Debt and Industry Product Markets: An Empirical Analysis

Contents

1 Big picture

- **Question.** Does a firm's capital structure affect product-market behavior—output, pricing, market shares, capacity, and managerial incentives?
- **Core idea.** Debt can change product-market conduct because it changes incentives and constraints. Highly levered firms may become less aggressive because cash flow is committed to debt service, or more aggressive because equityholders prefer risk and limited liability.
- **Empirical setting.** The paper studies four industries in the 1980s in which leading firms sharply increased financial leverage: fiberglass insulation, tractor trailers, polyethylene chemicals, and gypsum.
- **Why these industries.** They have relatively small numbers of firms, identifiable recapitalizations or LBOs, and usable product-level price and quantity data.
- **Main result.** In fiberglass, tractor trailers, and polyethylene, output is negatively associated with the industry debt ratio. Product prices and margins tend to rise, while recapitalizing firms lose or fail to gain market share.
- **Contrasting case.** In gypsum, output is positively associated with debt and price falls. This industry has lower entry barriers and important rivals that did not become highly levered.
- **Managerial incentives.** Compensation and ownership evidence suggests that recapitalizations strengthen executive incentives to maximize shareholder value.
- **Takeaway.** Product-market effects of debt are not mechanically one-directional. The effect depends on rivals' leverage, entry barriers, capacity conditions, and managerial incentives.

2 Data and measurement

2.1 Industries and recapitalizations

The analysis focuses on four product markets:

- **Fiberglass insulation and roofing.** Owens-Corning Fiberglas sharply increases leverage in 1986.
- **Tractor trailers.** Fruehauf recapitalizes in 1986, and several other major firms also undertake LBOs or recapitalizations.
- **Polyethylene chemicals.** Quantum Chemical recapitalizes in 1988; Union Carbide also increases leverage; Cain is formed through an LBO of DuPont plants.
- **Gypsum wallboard.** USG recapitalizes in 1988, while National Gypsum had an earlier LBO.

Interpretation. The paper deliberately chooses industries where the capital-structure change is large and visible, making it more plausible that product-market behavior can change in response.

2.2 Market-share data

- Firm market shares come from annual reports, industry segment data, R.L. Polk data for trailers, Modern Plastics capacity data for polyethylene, and Bureau of Mines/Census data for gypsum and fiberglass.
- Market shares are summarized for three years before and three years after the leading firm's recapitalization.
- The recapitalization year is excluded from pre/post averages.

Interpretation. Market-share movements provide a first diagnostic: if debt makes a firm less aggressive, its share should fall or fail to rise; if debt induces aggressive output expansion, its share may rise.

2.3 Monthly product-market data

- The paper uses monthly price and quantity series from 1980–1990.
- Product prices are deflated by the wholesale price index.
- Demand-shift variables include construction, industrial production, auto production, and shipments of manufactures, depending on industry.
- Input-price controls include wages, oil, electricity, natural gas, steel, aluminum, glass abrasives, and other product-specific inputs.
- Substitute-price proxies include gypsum board, truck and bus bodies, and wood sheeting.

Interpretation. The key advantage over simple accounting regressions is that product-level price and quantity data separate output behavior from input-cost changes.

2.4 Accounting and capacity evidence

- The paper computes sales growth, price-cost margins, and capital expenditures over sales for the major recapitalizing firms.
- Price-cost margins use the Lerner-index style measure:

$$\frac{S - COGS + \Delta Inventory}{S + \Delta Inventory}.$$

- Plant closings and capacity utilization data are used as supporting evidence that capacity is actually withdrawn or expanded.

Interpretation. Accounting margins are noisy because average costs may include fixed and sunk costs. The paper therefore treats them as corroborating evidence rather than the final product-market test.

2.5 Executive compensation data

- Executive compensation data come from proxy statements for Owens-Corning Fiberglas, Fruehauf, Quantum Chemical, and USG.
- The sample includes executives who served for at least five consecutive years during 1977–1992.
- The paper measures salary and bonus, contingent compensation, option grants, share ownership, and pay-performance sensitivity.

Interpretation. This evidence tests whether recapitalization changes managerial incentives, not only financial constraints. The paper argues that incentives become more equity-oriented after recapitalization.

3 Models and empirical specifications

3.1 General demand and cost framework

The general inverse demand function is

$$P_t = D(Q_t, Y_t, Z_t, \alpha, \varepsilon_t), \tag{1}$$

where P_t is price, Q_t is industry quantity, Y_t is a vector of demand shifters, and Z_t is a substitute price.

The firm cost function is

$$C_{it} = C(Q_{it}, W_t, \beta, \mu_{it}), \tag{2}$$

where W_t is a vector of input prices. Marginal cost is

$$MC_{it} = c(Q_{it}, W_t, \beta, \mu_{it}). \tag{3}$$

Interpretation. The model is an empirical industrial-organization framework: it connects pricing and quantity to demand shifts, input prices, and strategic conduct.

3.2 Supply relationship and conduct

The firm's supply condition can be written as

$$P_t = MC_{it} - \theta_i D_Q^{inv}(Q_t, Y_t, Z_t, \alpha, \varepsilon_t)Q_{it}. \tag{4}$$

- $\theta = 0$ corresponds to perfect competition: price equals marginal cost.
- $\theta = 1$ corresponds to joint monopoly behavior.
- Intermediate θ values capture oligopoly conduct.

Interpretation. Debt matters if it changes the conduct term or the perceived aggressiveness of firms in the product market.

3.3 Loglinear demand

The estimated loglinear demand equation is

$$q_t = \alpha_0 + \alpha_1 p_t + \alpha_2 y_t + \alpha_3 z_t + \varepsilon_t. \tag{5}$$

Interpretation. Demand shocks and substitute prices help identify the relationship between price and quantity and separate cost movements from demand movements.

3.4 Industry supply relationship with leverage

The industry supply relationship is estimated as

$$p_t = \beta_0 + \beta_1 q_t + \beta_j W_{jt} + \gamma DebtRatio_t + v_t. \tag{6}$$

- q_t is instrumented because price and quantity are jointly determined.
- $DebtRatio_t$ is the weighted-average debt ratio of public firms in the industry.
- A positive γ means price rises with leverage, conditional on quantity and input prices.
- A negative γ means price falls with leverage.

Interpretation. The sign of γ distinguishes whether debt is associated with less aggressive behavior and higher prices or with more aggressive output expansion and lower prices.

3.5 Reduced-form quantity equation

The reduced-form regressions use industry quantity as the dependent variable:

$$q_t = \delta_0 + \delta_1 DemandShifters_t + \delta_2 InputPrices_t + \delta_3 DebtRatio_t + u_t. \tag{7}$$

Interpretation. The reduced form asks directly whether industry output rises or falls when the industry debt ratio rises, controlling for the product-market environment.

3.6 Compensation regressions

The compensation tests regress changes in executive compensation on shareholder wealth, stock returns, operating performance, sales, and executive-position dummies. The paper estimates both a linear specification and a loglinear specification.

Interpretation. This tests whether recapitalization changes the objective function facing managers. If pay becomes more sensitive to shareholder wealth after recapitalization, managerial incentives become more aligned with owners.

4 Identification and empirical design

4.1 What identifies the product-market effect

- The key variation is the sharp change in financial leverage around recapitalizations and LBOs.
- Monthly price and quantity regressions use input prices and demand shifters to isolate changes in product-market conduct.
- Industry-specific evidence allows the paper to interpret signs using market structure: rival leverage, entry barriers, minimum efficient scale, and fixed costs.

Interpretation. The empirical strategy is not a randomized experiment. Its strength is triangulation: market shares, price/quantity regressions, accounting measures, capacity evidence, and compensation evidence all point to related mechanisms.

4.2 Why industry heterogeneity matters

- Fiberglass, trailers, and polyethylene have high fixed costs, high leverage among rivals, or difficult expansion by competitors.
- Gypsum has lower barriers to entry and important rivals that did not become highly levered.
- The sign of the leverage-output relation differs across these environments.

Interpretation. The paper's most important lesson is conditionality. Debt can soften competition in one setting and intensify it in another, depending on product-market structure.

4.3 Limits of the design

- Recapitalizations are endogenous corporate events.
- There are only four industries, so the paper relies on detailed case-style evidence rather than broad cross-sectional statistical power.
- Some industry data are proxies: capacity data substitute for sales in polyethylene, and accounting margins are imperfect measures of marginal cost.
- The exact contribution of debt versus other simultaneous restructuring actions cannot be perfectly separated.

Interpretation. The paper should be read as an influential empirical bridge between corporate finance and industrial organization rather than as a clean natural experiment.

5 Tables and figures

The paper has no standalone figures. The notes therefore group the original tables directly cropped from the paper. Long tables are split into adjacent panels to preserve readability.

5.1 Table 1: Market descriptions

Read:

- Owens-Corning's fiberglass share falls from 30.5% to 26.5% after its recapitalization, while the next two unleveraged firms rise from 22.1% to 25.7%.
- Fruehauf's tractor-trailer share falls from 22.4% to 16.5%; the top five leveraged firms fall from 48.1% to 41.6%, while unleveraged firms rise from 19.7% to 24.8%.
- In polyethylene, Quantum's share is roughly flat, Union Carbide falls, Cain rises, and unleveraged rivals rise.
- Gypsum is different: USG rises from 41.1% to 51.1%, and the next two unleveraged firms also rise.

Economic message. The descriptive evidence already shows that leverage does not mechanically produce market-share gains. In the first three industries, high leverage is associated with loss of share or failure to gain share; in gypsum, leverage coincides with expansion.

Interpretation. Table 1 is the paper's first product-market diagnostic. It links capital structure changes to concrete industry outcomes: who loses share, who gains share, and what entry barriers look like. The contrast between gypsum and the other industries is the central comparative case that motivates the rest of the paper.

Table 1

Market descriptions

Pre-recap figures are the means for three years prior to the leading firm's recapitalization announcement. Post-recap figures are means for three years after the recapitalization, or until the 1990 fiscal year-end. The exact years used in each of the pre- and post-recap numbers for each firm are given in the footnotes. Debt-to-value ratios are one-year pre- and post-recapitalization book value of debt divided by the book value of debt plus the market value of equity. MES stands for minimum efficient scale; low versus high is according to industry self-descriptions.

Industry	Three-year % industry sales: Pre-recap	Three-year % industry sales: Post-recap	Capital structure changes	Technology/ Barriers to entry
Fiberglass roofing and insulation				
Leading firm: Owens-Corning Fiberglas (OCF)[a]	30.5%	26.5%	OCF debt/value: Pre-recap = 34%, post = 71%	High MES Large fixed costs
2nd firm: Manville	22.3%	21.8%		
Manville before bankruptcy	27.1%			
Next 2 'unleveraged' firms	22.1%	25.7%		
Tractor trailers				
Leading firm: Fruehauf[b]	22.4%	16.5%	Fruehauf debt/value: Pre-recap = 22%, post = 72%	Job shop
Top 5 leveraged firms	48.1%	41.6%	Five of top six firms involved in leveraged recaps/LBOs.[b]	Distribution network
Top 5 'unleveraged' firms	19.7%	24.8%		
Polyethylene				
Leading firm: Quantum[c]	21.8%	21.9%	Quantum debt/value:	
2nd firm: Union Carbide[d]	14.2%	13.2%	Pre-recap = 36%, Post = 76%	High MES
3rd firm: Cain Inc.[d]	6.3%	7.9%	Union Carbide debt/value:	Large fixed costs
Top 4 'unleveraged' firms	32.1%	34.8%	Pre-recap = 37%, Post = 55%	
Gypsum				
Leading firm: USG[e]	47.7%	51.1%	USG debt/value:	Low MES
2nd firm: National Gypsum	22.1%	21.6%	Pre-recap = 35%, Post = 90%	Low fixed costs
Next 2 'unleveraged' firms	15.7%	22.1%	National Gypsum: 1984 LBO.	

Individual firm data are obtained from industry segment data from firm annual reports for fiberglass and gypsum. Total market size data are from the U.S. Bureau of Mines and the Census Department *Annual Survey of Manufactures* for fiberglass and gypsum. Data for tractor trailers are from R.L. Polk & Co. and represents actual unit quantity sales. Data for polyethylene represents rated capacity of plants and are from *Modern Plastics*, January annual issue.

[a]Owens–Corning Fiberglas' recapitalization was announced 8/28/86 and completed in November 1986. The three-year averages exclude 1986, representing 1983–1985 for the pre-recap period and 1987–1990 for the post-recap period. The market share for Owens–Corning Fiberglas rises to 35% before the recapitalization if some specialty fiberglass products that were sold off are included. The market share decrease would then rise to 8.5 percentage points if the market size of these products were included in the denominator.

[b]Fruehauf's recapitalization was announced 6/24/86 and completed in December 1986. The pre-recap period represents 1983–1985. The post-recap period represents 1987–1989. Four other firms (Dorsey, Great Dane, Monon, Trailmobile) recapitalized using LBOs from 1985 to early 1987. The Fruehauf pre- and post-recap periods are used for the market share numbers for the combined five firms.

[c]Quantum Chemical's recapitalization was announced 12/27/88 and completed in January 1989. The pre-recap period represents years 1986–1988. The post-recap period for Quantum Chemical only represents two years, 1989 and 1990, given that 1991 capacity data was not yet available.

[d]The pre-recap period for Union Carbide represents 1983–1985, three years before its recapitalization in 1986. Cain Inc. was formed by an LBO of DuPont high-density polyethlyene plants in 1987. Complicating identification of the post-recap period for Cain, Occidental Petroleum bought these plants nine months later. Thus the same years as Quantum Chemical are used. If the pre-recap period is defined as 1984–1986 for Cain, its market share number would be 5.92%.

[e]USG's recapitalization was announced in 5/2/88 and completed in July 1988. The pre-recap period represents 1985–1987. The post-recap period represents 1989–1990. The market share numbers for USG are overstated because USG has sales in some specialty gypsum products which are not separated from general gypsum sales, while the market size from the Bureau of the Mines does not include these products. This overstatement should primarily affect the level and not the change in market share.

5.2 Table 2: Industry product-market data

Read:

- Fiberglass insulation prices are roughly stable after recapitalization, while quantity rises to 1.18 and oil prices fall sharply.
- Tractor-trailer prices rise from 31.31 to 34.4, while quantity falls to 0.79.
- Polyethylene prices and quantities both rise, while oil prices fall.
- Gypsum prices fall to 0.75 while quantity rises to 1724 million square feet.

Economic message. Input-price movements alone cannot explain the observed product-price and quantity patterns. The table motivates the structural price/quantity regressions by showing that post-recapitalization outcomes differ sharply across industries.

Interpretation. Table 2 matters because it prevents a simple cost-story interpretation. In some cases, input prices fall while output or price moves in directions that require a product-market explanation. This makes the later supply regressions more credible and necessary.

Fruehauf's, Quantum Chemical's, and USG's announcements of leveraged recapitalizations, with exact years given in the footnotes. Unless otherwise indicated, the price and quantity series are indices. All indices are normalized for presentation purposes only so that the pre-recap mean equals 1.0. All price data are deflated by the wholesale price index.

Industry	Pre-recap mean	Std. error of mean	*N*	Post-recap mean	Std. error of mean	*N*
Fiberglass insulation[a]						
Insulation price (Index)	1.0	0.006	75	0.99	0.006	50
Quantity	1.0	0.009		1.18[e]	0.007	
Real input prices						
Glass	1.0	0.004		1.02[e]	0.006	
Wage: SIC 3296 ($)	8.60	0.054		9.28[e]	0.040	
Oil	1.0	0.024		0.57[e]	0.019	
Electricity	1.0	0.006		1.04[e]	0.006	
Tractor trailer[b]						
Price tank trailers ($000)	31.31	0.203	77	34.4 [e]	0.197	55
Quantity tank trailers	1.0	0.028		0.79[e]	0.013	
Real input prices						
Aluminum	1.0	0.032		1.16[e]	0.029	
Steel	1.0	0.003		1.02[e]	0.006	
Wage: SIC 3215 ($)	8.28	0.068		8.74[e]	0.060	
Electricity	1.0	0.009		1.05[e]	0.007	
Polyethylene chemical[c]						
High-density poly. price	1.0	0.015	97	1.17[e]	0.031	25
Quantity (mil. lbs)	452.0	10.778		638.2 [e]	12.027	
Low-density poly. price	1.0	0.002		1.14[e]	0.039	
Quantity (mil. lbs)	620.2	9.49		709.3 [e]	23.930	
Real input prices						
Oil	1.0	0.029		0.71[e]	0.038	
Wage: SIC 2821 ($)	11.66	0.120		12.82[e]	0.034	

Table 2 (continued)

Industry	Pre-recap mean	Std. error of mean	*N*	Post-recap mean	Std. error of mean	*N*
Gypsum[d]						
Gypsum board price	1.0	0.013	100	0.75[e]	0.009	29
Quantity (mil. sq. feet)	1432	26.960		1724[e]	21.160	
Real input prices						
Electricity	1.0	0.008		1.01	0.007	
Natural gas	1.0	0.014		0.73[e]	0.006	
Wage: SIC 327 ($)	9.02	0.056		9.38[e]	0.027	
Paper sheeting	1.0	0.003		1.69[e]	0.018	

[a]The pre-recap period for the fiberglass industry is from April 1980 to July 1986, the months before Owens–Corning Fiberglas' recapitalization announcement on 8/28/86. The post-recap period is from August 1986 to October 1990.

[b]The pre-recap period for the tractor trailer industry is from January 1980 to May 1986, the months before Fruehauf's recapitalization announcement on 6/24/86. The post-recap period is from June 1986 to December 1990.

[c]The pre-recap period for the polyethylene industry is from November 1980 to November 1988, the months before Quantum Chemical's recapitalization announcement on 12/27/88. The post-recap period is from December 1988 to December 1990.

[d]The pre-recap period for the gypsum industry is from January 1980 to April 1988, the months before USG's recapitalization announcement on 5/2/88. The post-recap period is from May 1988 to September 1990.

5.3 Table 3: Accounting measures of performance

Read:

- Owens-Corning's own sales fall in the first post-recapitalization year, while its price-cost margin rises from the pre-period into years 1–3.
- Fruehauf's sales fall strongly in year 1, while capital expenditures over sales decline.
- Quantum Chemical's margin rises after recapitalization, although acquisition and industry effects complicate interpretation.
- USG is different: its price-cost margin falls after recapitalization, consistent with more aggressive output behavior in gypsum.

Economic message. The accounting results reinforce the product-market evidence. In the first three industries, high leverage is associated with higher margins and lower sales/capex; in gypsum, margins fall.

Interpretation. Table 3 should not be read as a clean marginal-cost test, because accounting data mix fixed costs, sunk costs, and input-cost timing. Its value is corroborative: it aligns firm-level accounting behavior with the direction of industry price/quantity movements.

5.4 Tables 4a–4d: Product-market equations by industry

Read:

- Tables 4a–4c show that in fiberglass, tractor trailers, and polyethylene, the debt ratio is associated with higher prices or lower output in the product-market system.

Table 3

Accounting measures of performance

Common measures of firm performance using accounting data. Years are relative to the firm's announcement of a recapitalization. Year 0 is the year of the announcement. The announcement date is given under the firm's name. Thus, for Owens–Corning Fiberglas, years 1–3 are 1987–1989. All data are obtained from firm annual reports. Industry numbers represent industry medians for at least three firms at the four- or three-digit SIC code. Industry data are obtained from COMPUSTAT. Sales change is the annual percentage change (not multiplied by 100). Operating price–cost operating margin is sales − cost of goods sold + change in inventories divided by sales + change in inventories. This margin is also called the Lerner Index.

		Year − 2	Year − 1	Year 1	Year 2	Year 3	Wilcoxon test Before vs. after: *p*-value[a]
Owens–Corning Fiberglas (announcement date: 8/28/86)							
Sales change	Own	0.097	0.094	− 0.125	− 0.021	0.060	0.05
	Industry (N = 12)	0.090	0.010	0.156	0.068	0.034	
Price–cost margin (Lerner index)	Own	0.322	0.328	0.335	0.381	0.381	0.02
	Industry	0.300	0.290	0.351	0.344	0.343	
Capital expend./Sales	Own	0.043	0.059	0.034	0.045	0.042	0.06
	Industry	0.053	0.065	0.049	0.063	0.041	
Fruehauf Corporation[b] (announcement date: 6/24/86)							
Sales change	Own	− 0.080	0.047	− 0.269	0.068		0.06
	Industry (N = 12)	− 0.013	0.020	0.017	0.085		
Price–cost margin (Lerner index)	Own	0.143	0.083	0.190	0.161		0.20
	Industry	0.397	0.272	0.258	0.184		
Capital expend./Sales	Own	0.039	0.031	0.028	0.025		0.02
	Industry	0.035	0.043	0.030	0.019		
Quantum Chemical[b] (announcement date: 12/27/88)							
Sales change	Own	0.113	0.554	− 0.086	− 0.006		0.15
	Industry (N = 10)	0.196	0.151	0.045	0.033		
Price–cost margin (Lerner index)	Own	0.173	0.220	0.311	0.255		0.06
	Industry	0.339	0.384	0.378	0.366		
Capital expend./Sales	Own	0.070	0.068	0.221	0.175		0.02
	Industry	0.057	0.075	0.090	0.099		
USG Corporation (announcement date: 5/2/88)							
Sales change	Own	0.078	0.064	− 0.224	− 0.025	− 0.126	0.48
	Industry (N = 7)	0.035	0.068	0.04 5	0.117	0.067	
Price–cost margin (Lerner index)	Own	0.376	0.341	0.290	0.309	0.272	0.00
	Industry	0.322	0.336	0.304	0.307	0.305	
Capital expend./Sales	Own	0.072	0.062	0.038	0.036	0.033	0.61
	Industry	0.041	0.038	0.059	0.042	0.075	

[a]Observations are separated into before and after sets. The year of the recapitalization is excluded. The Wilcoxon signed-rank test is a nonparametric test. The null hypothesis is that the before and after measures are from populations with the same distribution and the same medians. The Wilcoxon signed-rank test for the own row includes *all* firms in the industry. Results were similar in tests excluding the recapitalizing firm. Separate tests for firm versus the industry are not conducted because of the limited number of observations.

[b]Year 3 is unavailable for Fruehauf because it was acquired by another corporation in the third year following the recapitalization. Year 3 was unavailable for Quantum because of the late date of its recapitalization.

- The reduced-form quantity regressions in those industries generally show a negative relation between output and average industry debt.
- Table 4d for gypsum reverses the pattern: output is positively associated with debt, and the price relation differs from the other industries.
- The regressions control for demand shifters, input prices, substitute prices, seasonality, and demand shocks.

Economic message. The central empirical result is in Tables 4a–4d: leverage interacts with product-market conduct. The sign differs by industry structure, so debt is not simply "strategic aggression" or "strategic retreat" in all settings.

Interpretation. These tables are the paper's main bridge between corporate finance and IO. The debt coefficient is not just a financing variable; it is interpreted as shifting the industry supply relationship. The more levered industries with barriers and levered rivals behave as if debt softens output competition. Gypsum behaves as if leverage supports output expansion because low entry barriers and less-levered rivals change strategic incentives.

Table 4a

Industry product market analysis: Fiberglass insulation industry

Supply relationships and the reduced-form equation test whether capital structure interacts with industry product markets. The linear (loglinear) form of the industry demand equation in the first panel is estimated with the linear (loglinear) supply relationship in the second panel using two-stage simultaneous equations. The before/after rows for the loglinear supply relationship are estimated in one equation. The before/after periods are months in the 1980's surrounding Owens–Corning Fiberglas' announcement of a leveraged recapitalization on 8/28/86. The reduced-form of the loglinear demand and supply equations in the last panel is estimated in one step. Figures in parentheses are Newey–West heteroskedastic and autocorrelation consistent standard errors.

Demand equations – Dependent variable: Quantity

	Price	Construction	Industrial production	Auto production	Relative price[a] (Gypsum board)	R^2
Linear	− 0.81[e]	0.01[e]	1.04[e]	0.22[e]	0.02	0.87
	(0.11)	(0.002)	(0.21)	(0.08)	(0.03)	
Loglinear	− 0.96[e]	0.32[e]	1.15[e]	− 0.050	0.09[e]	0.86
	(0.10)	(0.06)	(0.20)	(0.06)	(0.03)	

Supply relationships – Dependent variable: Price

	Quantity	Electricity price	Glass abrasives price	Oil price	Wage	Demand shock[b]	Debt ratio[c]	Financial change dummy[d]	R^2
Linear	0.071[e]	0.37[e]	− 0.12	0.03	0.37[e]	0.19[e]	0.013[f]		0.90
	(0.02)	(0.05)	(0.15)	(0.02)	(0.08)	(0.07)	(0.01)		
Loglinear	0.00	0.27[e]	− 0.54[e]	0.01	0.60[e]	0.19[e]	0.021[e]		0.93
	(0.30)	(0.04)	(0.12)	(0.01)	(0.06)	(0.05)	(0.01)		
Loglinear									
Before	− 0.15[e]	0.30[e]	0.29[f]	0.00	0.21[e]	0.15[e]		0.147[e]	0.93
	(0.04)	(0.04)	(0.17)	(0.01)	(0.07)	(0.07)		(0.03)	
After	− 0.33[e]	0.21[e]	− 0.31	− 0.01	− 0.24	0.06			
	(0.09)	(0.07)	(0.24)	(0.02)	(0.29)	(0.04)			

Reduced form – Dependent variable: Quantity

	Construction	Industrial production	Auto production	Relative price[a]	Electricity	Glass abrasives price	Oil price	Wage	Debt ratio[c]	R^2
Loglinear	0.32[e]	0.82[e]	− 0.24[e]	0.02	− 0.86[e]	0.88[e]	− 0.24[e]	− 0.51[e]	− 0.09[e]	0.94
	(0.07)	(0.18)	(0.11)	(0.04)	(0.11)	(0.09)	(0.05)	(0.18)	(0.03)	

Data are monthly indices from 4/1980 to 10/1990. Price data were obtained from the Bureau of Labor and Statistics. Insulation quantity data and general demand shift data were obtained from the Federal Reserve Board's *Industrial Production*. Instruments for price in the demand equation are the omitted input prices. In the supply relationships, instruments for quantity and the quantity interaction are the demand variables and the omitted relative prices. To account for seasonality in demand, monthly dummies were included but are not reported. A time trend was also included and all price data are deflated by the wholesale price index.

[a]In the linear case the relative price is the price of gypsum board divided by the the price of fiberglass. In the loglinear case the relative price is the price of gypsum board.

[b]The demand shock variable is the residuals from the demand equation.

[c]Debt ratio is the weighted-average debt-to-market value ratio of the public firms, weighted by sales. In the linear case, the debt ratio variable is interacted with instrumented quantity divided by the substitute price.

[d]The financial change dummy variable equals one for the months following the capital structure change.

[e]Significant at the 5% level using a two-tailed test.

[f]Significant at the 10% level using a two-tailed test.

Table 4a: Fiberglass insulation.

Table 4b: Tractor trailers.

Table 4c: Polyethylene.

Table 4d: Gypsum.

Table 4b
Industry product market analysis: Tractor trailer industry (tank trailers)

Supply relationships and the reduced-form equation test whether capital structure interacts with industry product markets. The linear (loglinear) form of the industry demand equation in the first panel is estimated with the linear (loglinear) supply relationship in the second panel using two-stage simultaneous equations. The before/after rows for the loglinear supply relationship are estimated in one equation. Before and after periods are months in the 1980's surrounding Fruehauf's announcement of a leveraged recapitalization on 6/24/86. The reduced-form of the loglinear demand and supply equations in the last panel is estimated in one step. Figures in parentheses are Newey–West heteroskedastic and autocorrelation consistent standard errors.

Demand equations – Dependent variable: Quantity

	Price	Construction	Shipments of manufactures	Relative price[a] (truck and bus)	R^2
Linear	−0.60[e]	0.18	3.54[e]	0.16	0.45
	(0.31)	(0.12)	(0.50)	(0.20)	
Loglinear	−0.71[e]	1.02[e]	3.81[e]	0.34	0.54
	(0.39)	(0.33)	(0.50)	(0.35)	

Supply relationships – Dependent variable: Price

	Quantity	Steel price	Electricity price	Aluminum price	Wage	Demand shock[b]	Debt ratio[c]	Financial change dummy[d]	R^2
Linear	0.26	−0.24	1.35[e]	0.15[f]	0.10	−0.07	0.41[e]		0.58
	(0.27)	(0.35)	(0.59)	(0.09)	(0.44)	(0.10)	(0.16)		
Loglinear	0.23	−0.17	1.54[e]	0.104[f]	0.00	−0.04	0.203[e]		0.65
	(0.16)	(0.30)	(0.70)	(0.05)	(0.08)	(0.04)	(0.10)		
Loglinear									
Before	−0.18	0.56	1.09[e]	0.03	−0.43	−0.02		0.359[e]	0.61
	(0.18)	(0.84)	(0.46)	(0.08)	(0.48)	(0.02)		(0.09)	
After	0.27[e]	0.77[e]	0.31	−0.18[e]	−0.41	−0.07			
	(0.09)	(0.30)	(0.36)	(0.07)	(0.46)	(0.08)			

Reduced form – Dependent variable: Quantity

	Construction	Shipments of manufactures	Relative price[a]	Steel price	Electricity price	Aluminum price	Wage	Debt ratio[c]	R^2
Loglinear	0.24	1.42[e]	0.15	−0.02	1.16[e]	0.02	−0.17	−0.17[e]	0.58
	(0.30)	(0.34)	(0.20)	(0.38)	(0.45)	(0.10)	(0.58)	(0.06)	

Data are monthly indices from January 1980 to December 1990. Tractor quantity data were obtained from R.L. Polk. Tractor price data were obtained from *U.S. Department of Commerce: Current Industrial Reports*. All other price data were obtained from the Bureau of Labor and Statistics. Demand shift data, construction and shipments of manufactures, were obtained from the Federal Reserve *Industrial Production*. Instruments for price in the demand equation are the omitted input prices. In the supply relationships instruments for quantity and the quantity interaction are the demand variables and the omitted relative prices. To account for seasonality in demand, monthly dummies were included but are not reported. A time trend was also included and all price data are deflated by the wholesale price index.

[a]In the linear case the relative price is the price of small truck and bus bodies divided by the the price of tractor trailers. In the loglinear case the relative price is the price of tank tractor trailers.

[b]The demand shock variable is the residuals from the demand equation.

[c]Debt ratio is the weighted-average debt-to-market value ratio of the public firms, weighted by sales. In the linear case, the debt ratio variable is interacted with instrumented quantity divided by the substitute price.

[d]The financial change dummy variable equals one for the months following the capital structure change.

[e]Significant at the 5% level using a two-tailed test.

[f]Significant at the 10% level using a two-tailed test.

Table 4c
Industry product market analysis: Polyethylene industry (high-density polyethylene)

Supply relationships and the reduced-form equation test whether capital structure interacts with industry product markets. The linear (loglinear) form of the industry demand equation in the first panel is estimated with the linear (loglinear) supply relationship in the second panel using two-stage simultaneous equations. The before/after rows for the loglinear supply relationship are estimated in one equation. Before and after periods are months in the 1980's surrounding Quantum Chemical's announcement of a leveraged recapitalization on 12/27/88. The reduced-form of the loglinear demand and supply equations in the last panel is estimated in one step. Figures in parentheses are Newey–West heteroskedastic and autocorrelation consistent standard errors.

Demand equations – Dependent variable: Quantity

	Price	Consumer industrial production	Auto production	Shipments of manufactures	Relative price[a] (aluminum cans)	R^2
Linear	−0.71[e]	2.22[e]	0.11	−0.21	0.48[e]	0.91
	(0.24)	(0.96)	(0.16)	(0.33)	(0.72)	
Loglinear	−0.45[e]	1.19[f]	0.19[e]	−0.28	0.26	0.91
	(0.20)	(0.65)	(0.10)	(0.37)	(0.17)	

Supply relationships – Dependent variable: Price

	Quantity	Oil price	Electricity price	Wage	Demand shock[b]	Debt ratio[c]	Financial change dummy[d]	R^2
Linear	−1.12[e]	−0.47[e]	−0.84	0.22	0.08	0.33[e]		0.71
	(0.44)	(0.16)	(0.58)	(0.96)	(0.10)	(0.15)		
Loglinear	0.15	−0.23[e]	−1.23[e]	−0.64	0.12	0.15[f]		0.65
	(0.17)	(0.09)	(0.49)	(0.58)	(0.14)	(0.08)		
Loglinear								
Before	0.46[e]	−0.21[e]	−1.52[e]	−0.45[e]	−0.10		0.94[e]	0.64
	(0.24)	(0.05)	(0.60)	(0.74)	(0.12)		(0.39)	
After	0.20	−0.56[e]	1.91[f]	4.66[e]	0.47			
	(0.16)	(0.22)	(1.12)	(1.37)	(0.84)			

Reduced form – Dependent variable: Quantity

	Consumer industrial production	Auto production	Shipments of manufactures	Relative price[a]	Oil price	Electricity price	Wage	Debt ratio[c]	R^2
Loglinear	0.27	−0.10	0.89[f]	0.17	0.03	−0.85[e]	1.66[e]	−0.19[f]	0.92
	(0.81)	(0.14)	(0.47)	(0.33)	(0.05)	(0.41)	(0.64)	(0.10)	

Data are monthly indices from October 1980 to December 1990. Polyethylene quantity data were obtained from McGraw Hill/Data Resources Institute (DRI). Price data were obtained from the Bureau of Labor and Statistics. Demand shift data, consumer industrial and auto production, were obtained from the Federal Reserve Board's *Industrial Production*. Instruments for price in the demand equation are the omitted input prices. In the supply relationships instruments for quantity and the quantity interaction are the demand variables and the omitted relative prices. To account for seasonality in demand, monthly dummies were included but are not reported. A time trend was also included and all price data are deflated by the wholesale price index.

[a]In the linear case the relative price is the price of aluminum cans divided by the price of polyethylene. In the loglinear case the relative price is the price of aluminum cans.

[b]The demand shock variable is the residuals from the demand equation.

[c]Debt ratio is the weighted-average debt-to-market value ratio of the public firms, weighted by sales. In the linear case, the debt ratio variable is interacted with instrumented quantity divided by the substitute price.

[d]The financial change dummy variable equals one for the months following the capital structure change.

[e]Significant at the 5% level using a two-tailed test.

[f]Significant at the 10% level using a two-tailed test.

Table 4d
Industry product market analysis: Gypsum industry

Supply relationships and the reduced-form equation test whether capital structure interacts with industry product markets. The linear (loglinear) form of the industry demand equation in the first panel is estimated with the linear (loglinear) supply relationship in the second panel using two-stage simultaneous equations. The before/after rows for the loglinear supply relationship are estimated in one equation. Before and after periods are months in the 1980's surrounding USG's announcement of a leveraged recapitalization on 5/2/88. The reduced-form of the loglinear demand and supply equations in the last panel is estimated in one step. Figures in parentheses are Newey–West heteroskedastic and autocorrelation consistent standard errors.

Demand equations – Dependent variable: Quantity

	Price	Construction	Shipments of manufactures	Relative price[a] (wood sheeting)	R^2
Linear	0.44[e]	0.75[e]	−0.09	−0.07	0.89
	(0.20)	(0.12)	(0.13)	(0.05)	
Loglinear	−0.20	1.16[e]	−0.52[e]	−0.08	0.89
	(0.26)	(0.19)	(0.22)	(0.08)	

Supply relationships – Dependent variable: Price

	Quantity	Gas price	Electricity price	Wage	Demand shock[b]	Debt ratio[c]	Financial change dummy[d]	R^2
Linear	−0.05	0.72	1.19[f]	0.02	−0.06	−0.25[e]		0.65
	(0.13)	(0.47)	(0.64)	(0.01)	(0.18)	(0.07)		
Loglinear	0.52[e]	0.30	0.45	0.49	−0.16	−0.08[e]		0.61
	(0.26)	(0.19)	(1.00)	(0.19)	(0.14)	(0.04)		
Loglinear								
Before	1.04[e]	0.05	0.81[e]	−0.11	0.04		−0.12[f]	0.93
	(0.11)	(0.03)	(0.31)	(0.09)	(0.08)		(0.07)	
After	0.59[e]	−0.16	0.07	0.67[f]	−0.02			
	(0.12)	(0.12)	(0.16)	(0.40)	(0.06)			

Reduced form – Dependent variable: Quantity

	Construction	Shipments of manufactures	Relative price[a] (wood sheeting)	Gas price	Electricity price	Wage	Debt ratio[c]	R^2
Loglinear	0.72[e]	0.42[e]	0.07	0.12[e]	0.65[e]	1.47[e]	0.02[f]	0.94
	(0.06)	(0.19)	(0.05)	(0.04)	(0.16)	(0.28)	(0.01)	

Data are monthly indices from January 1980 to September 1990. Gypsum price and quantity data were obtained from Bureau of the Mines. All other price data were obtained from the Bureau of Labor and Statistics. Demand shift data were obtained from the Federal Reserve Board's *Industrial Production*. Instruments for price in the demand equation are the omitted input prices. In the supply relationships instruments for quantity and the quantity interaction are the demand variables and the omitted relative prices. To account for seasonality in demand, monthly dummies were included but are not reported. A time trend was also included and all price data are deflated by the wholesale price index.

[a]In the linear case the relative price is the price of wood sheeting divided by the the price of gypsum. In the loglinear case the relative price is the price of wood sheeting.

[b]The demand shock variable is the residuals from the demand equation.

[c]Debt ratio is the weighted-average debt-to-market value ratio of the public firms, weighted by sales. In the linear case, the debt ratio variable is interacted with instrumented quantity divided by the substitute price.

[d]The financial change dummy variable equals one for the months following the capital structure change.

[e]Significant at the 5% level using a two-tailed test.

[f]Significant at the 10% level using a two-tailed test.

5.5 Table 5 and Table 6: Managerial incentives and compensation

Read:

- Table 5 shows that contingent compensation and share ownership increase for many executives after recapitalization.
- Officers' and directors' share ownership rises sharply in several firms, especially Fruehauf.
- Table 6 shows that pay becomes more closely linked to shareholder wealth after recapitalization.
- Before recapitalization, changes in sales are more strongly connected to compensation; after recapitalization, shareholder wealth becomes more central.

Economic message. Recapitalization is not only a balance-sheet event. It changes managerial incentives, shifting attention from sales expansion toward shareholder value and cash-flow discipline.

Interpretation. These tables connect the product-market evidence to agency theory. If managers previously pursued sales growth or empire building, the recapitalization gives them both constraints and incentives to reduce unprofitable expansion. This is why the paper interprets some output reductions as potentially efficiency-enhancing rather than simply anticompetitive withdrawal.

Table 5

Management compensation surrounding recapitalizations

Management compensation data are summary statistics for 27 executives that served a minimum of five consecutive years during 1977–1992. All monetary data are in CPI-deflated 1983 dollars. Compensation data are from yearly proxy statements. Share ownership data are from the recapitalization prospectuses and represent fully-diluted ownership positions.

	Before/After recapitalization	Number of executives[a]	Mean salary and bonus	Mean contingent compensation[c]	Mean value stock options[d]	Officers' and directors' share ownership
Owens–Corning Fiberglas	Before	6	288,053	21,712	39,846	1.1%
	After	5	615,710	109,416	173,650	6.1%
Fruehauf[b]	Before	5	327,635	0	37,697	0.7%
	After	2	267,829	0	36,071	28.7%
Quantum Chemical	Before	6	307,050	55,673	81,785	3.3%
	After	3	401,655	188,520	359,730	4.7%
USG	Before	7	382,294	55,507	106,515	1.1%
	After	5	403,451	165,243	210,146	3.6%

Data are for 15 years surrounding the recapitalizations excluding the year of the recapitalization announcement. Data for Owens–Corning Fiberglas' pre-recapitalization period are for the years 1977–1985. Data for the post-recapitalization period are for the years 1987–1992. Similarly for Fruehauf, the pre-recapitalization years are 1977–1985 and the post-recapitalization years are 1987–1989. For Quantum Chemical, the pre-recapitalization years are 1977–1987 and the post-recapitalization years are 1989–1992. For USG, the pre-recapitalization years are 1977–1987 and the post-recapitalization years are 1989 1992.

[a]An executive can be in both the before and after periods.

[b]Fruehauf compensation data ends in 1989 when it was acquired by another firm.

[c]Any stock grants are included in contingent compensation and are valued at the time of the grant. Changes in the value of stockholdings are not included in the compensation variables.

[d]Stock options are valued using the Black–Scholes formula. The last 60 months of stock return data are used to compute the estimated standard deviation of stock returns.

Table 6

Management compensation and shareholder wealth

Regressions test whether changes in management compensation are related to changes in shareholder wealth and sales. The first panel contains the linear estimation. In the loglinear relationship in the second panel, coefficients represent elasticities of management compensation with respect to the independent variables. Coefficients are estimated with a separate intercept (fixed effect) for each executive and for each firm-year. Executives have served a minimum of five years over 1977–1992. The chairman of the board was also the CEO for all four companies. Standard errors are in parentheses.

Linear specification (Jensen–Murphy, 1990) – Dependent variable: Change in total compensation[a] (in 1983 $000)

	Change in shareholder wealth[b]			Position dummies		
	Current year	Lagged year	Change in sales	Chair./CEO	President	$\bar{R}^2$
Full period: 1977–1992						
Execs = 29	0.000429[e]	− 0.000115	0.0000402[e]	335.21	73.767	0.635
Exec. years = 201	(0.0000547)	(0.000063)	(0.0000104)	(211.42)	(171.57)	
Before recapitalization						
Execs = 24	0.0000643	− 0.000067	0.000134[e]	133.95	− 175.95[e]	0.214
Exec. years = 152	(0.0000549)	(0.000057)	(0.0000326)	(105.23)	(87.80)	
After recapitalization						
Execs = 15	0.000571[e]	− 0.000232	0.000195	542.621	− 45.943	0.545
Exec. years = 49	(0.000167)	(0.000185)	(0.000423)	(193.04)	(855.45)	

Loglinear specification (Murphy, 1985) – Dependent variable: Change in log total compensation[a] (in 1983 $000)

				Position dummies		
	Stock return[c]	Change in ln(Sales)	Return on assets[d]	Chair./CEO	President	$\bar{R}^2$
Full period: 1977–1992						
Execs = 29	0.252[e]	0.182	− 0.069	0.191	− 0.041	0.453
Exec. years = 201	(0.104)	(0.214)	(0.285)	(0.132)	(0.073)	
Before recapitalization						
Execs = 24	0.058	0.544[f]	0.098	0.402[f]	− 0.017	0.590
Exec. years = 152	(0.128)	(0.310)	(0.822)	(0.217)	(0.174)	
After recapitalization						
Execs = 15	0.493[e]	− 0.740	0.508	0.129	− 0.321	0.544
Exec. years = 49	(0.212)	(1.13)	(1.35)	(0.392)	(0.417)	

Chow structural change tests reject at the 1% level the hypothesis that the coefficients on shareholder wealth, stock return, and change in sales are the same for the before and after periods. *F*-statistics for the Chow tests were 6.12 for the linear and 6.45 for the loglinear regressions.

[a]Total compensation is defined as the sum of salary, bonus, contingent, and option compensation. Similar results are obtained, available from the author, without including the value of option grants. Any stock grants are included in contingent compensation and are valued at the time of the grant. The relationship is also estimated including changes in the value of existing stockholdings. Results discussed in the text, also available from the author, show that the sensitivity of pay to performance increases following the recapitalization. All compensation data are deflated by the consumer price index.

[b]Change in shareholder wealth is the change in the aggregate value, deflated by the consumer price index, of common stock including dividends for the current year and the prior year ending in December.

[c]Stock return is the return including any dividends for the year ending in December.

[d]Return on assets is the operating cash flow, before any interest payments, divided by the year-end book value of the firm's assets.

[e]Significant at the 5% level using a two-tailed t test.

[f]Significant at the 10% level using a two-tailed t test.

6 Short summary

- Phillips studies how increases in financial leverage affect product-market behavior in four detailed industries.
- The core empirical challenge is that capital structure and product-market outcomes are jointly determined.
- The paper addresses this by combining market shares, product-level price and quantity data, accounting measures, plant closings, and compensation data.
- In fiberglass, tractor trailers, and polyethylene, leverage is associated with lower output or less aggressive behavior.
- In gypsum, leverage is associated with higher output and lower price.
- This difference is linked to entry barriers, minimum efficient scale, fixed costs, and whether rival firms are also highly levered.
- Accounting evidence supports the product-market results: margins rise and sales/capex fall for many recapitalizing firms in the first three industries.
- Compensation evidence suggests that recapitalization increases executive alignment with shareholder wealth.
- The overall conclusion is that capital structure matters for product markets, but the effect depends on industry structure.

7 Conclusion

7.1 Core conclusion

Phillips shows that debt can influence product-market outcomes. In three of the four industries, higher leverage is associated with reduced output, higher margins, or market-share losses for recapitalizing firms. In the gypsum industry, however, the sign reverses: output rises and price falls. The paper therefore rejects a one-size-fits-all view of debt as either always softening or always intensifying competition.

7.2 Economic intuition

Debt changes incentives and constraints. A highly levered firm may need to conserve cash, reduce investment, close plants, and avoid aggressive output expansion. This can raise prices and margins if rivals are also constrained or if entry is hard. But when rivals are less levered and entry is easier, a recapitalized firm may use its new incentives to expand output or compete more aggressively. The same financial instrument can therefore generate different product-market outcomes depending on market structure.

7.3 Agency-cost interpretation

A central contribution of the paper is to connect product-market behavior with agency costs. Debt may force managers to disgorge cash flow, cut unproductive investment, and focus on shareholder value. The compensation evidence strengthens this interpretation: after recapitalizations, pay appears more tied to shareholder wealth. Output reductions in fiberglass, trailers, and polyethylene may therefore represent a move away from inefficient expansion rather than purely a reduction in competitive pressure.

7.4 Strategic-interaction interpretation

The paper also speaks to strategic capital-structure models. In Brander–Lewis style models, debt can make equityholders prefer riskier, more aggressive output choices. In Glazer/Phillips-style mechanisms, debt can commit firms to less aggressive behavior by increasing financial pressure. Phillips's evidence suggests that both mechanisms can be relevant, but they apply in different competitive environments. The same capital-structure change has a different strategic meaning when rivals are levered, capacity is costly, and entry barriers are high.

7.5 Implications for corporate finance

The paper broadens the role of capital structure. Debt is not merely a tax shield, a bankruptcy-cost tradeoff, or a governance device. It can affect real product-market choices. This means that financing decisions can change not only the distribution of cash flows among claimants but also the level and timing of production, investment, and capacity.

7.6 Implications for industrial organization

The paper also broadens IO analysis. Product-market conduct cannot be fully understood without firm financing. A firm with the same technology and market share may behave differently if it is highly levered. Therefore, ownership and capital structure are part of market conduct, especially in industries with concentrated firms and lumpy capacity.

7.7 Policy and antitrust implications

The results imply that antitrust and competition policy should pay attention to major recapitalizations and LBO waves, especially in concentrated industries. If many rivals become highly levered simultaneously, competition may soften through capacity reductions or lower output. But if leverage mainly disciplines inefficient incumbents or triggers more aggressive competition in low-barrier markets, the welfare effect can differ. The policy implication is not that leverage is bad, but that financial structure can be relevant to product-market outcomes.

7.8 Limitations and research agenda

The paper studies four industries in depth, so external validity remains an open question. Future work can use broader data to test whether the same patterns hold across many industries, distinguish agency-cost reductions from market-power effects, and examine whether debt-driven output reductions improve or harm welfare. Modern datasets with establishment-level production, transaction-level prices, and security-level financing could extend the paper's logic much further.

7.9 Takeaway

The enduring lesson is that corporate finance and product-market strategy are inseparable. A firm's debt choice can alter incentives, capacity, output, pricing, and managerial behavior. To understand competition, we must understand financing; to understand financing, we must understand the product market in which the firm operates.